TECHNICAL RETROSPECTIVE / 01

Reverse-engineering a Blotto field

Learning an opponent population from leaderboard scores.

THE CENTRAL IDEA

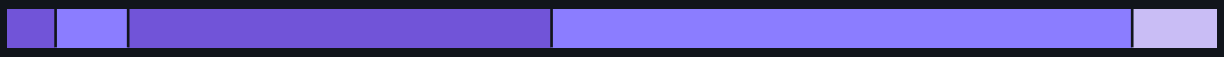
Each score is a measurement, not just a result.

I used inverse linear programming, exact dynamic programming, and decision-focused experiments to learn which allocations were worth testing.

INSIDE THE METHOD

01 Scores as linear constraints
02 A joint field, generated by columns
03 Probes chosen to reduce regret
04 Calibration, drift, and caveats

BEST-TESTED ALLOCATION / CASTLES 1 TO 5



(4, 6, 35, 48, 7)

100 SOLDIERS / 15 POINTS

OBSERVED PEAK RANK

3 / 4,153 entries

LAST SAVED SCORE

9.213 / 15 points

EXPERIMENT WINDOW

Oct 31–Nov 3, 2026

An independent study of Sequoia's "No Limit" Castle Challenge. Observed rankings are not a claim about final official standings. The final form receipt was not verified as a leaderboard replacement.

1 A leaderboard as an instrument

I did not need to see my opponents' bids to learn something useful about them. I needed to understand what a score measured.

I treated the Castle Challenge as a system-identification problem: which hidden opponent fields were consistent with my measurements, and which next experiment would change my decision? My best-tested split was $(4, 6, 35, 48, 7)$. These checkpoints put the result in context.[1,2]

CHECKPOINT / PACIFIC TIME	SCORE	RANK
#71: Nov 1, 3:07 PM PST	9.100	3 / 3,585
Re-test: Nov 3, 7:51 AM PST	9.094	3 / 4,153
Last saved board: Nov 3, 11:21 PM PST	9.213	7 / 4,784

On the last saved board, the leaders displayed 9.220: a gap of 0.007 points. This was a close observed result, not a proof of global optimality or a first-place finish.

The game and the measurement

Game 1 is head-to-head Colonel Blotto. I distribute 100 integer soldiers across five castles worth $1, \ldots, 5$ points. More soldiers wins a castle; a tie splits its points. The objective is average castle points against the field, not the probability of winning the entire match.

$$\mathcal{A} = \left\{ a \in \mathbb{Z}^5_{\geq 0} : \sum_{c=1}^{5} a_c = 100 \right\}. \tag{1}$$

Let X be a random opponent allocation, $P_c(k) = \Pr(X_c = k)$, and $F_c(k) = \Pr(X_c \leq k)$. Linearity of expectation gives the reduction that drove the entire method:

$$S(a) = \sum_{c=1}^{5} c \left[F_c(a_c - 1) + \frac{1}{2} P_c(a_c) \right]. \tag{2}$$

No independence assumption across castles is needed. Every score is a known linear functional of five unknown marginal distributions. A submission is therefore an experiment on the field, not merely another guess at a good allocation.[3]

Receipt is not reflection. I restored $(4, 6, 35, 48, 7)$ before the Nov 3, 11:59 PM PST deadline. The form confirmed receipt, but the saved boards never verified that final replacement. The 9.213 score belongs to an earlier reflected entry using the same split.

Finite-field detail: a self-match contributes 7.5 points. The early synthetic-population fitter subtracts 7.5 from the score total, then divides by N-1 to exclude the self-match. The joint LP uses field expectations, tolerances, and residual slack.

2

From measurements to a joint field

A plausible fitted population was a starting point. A set of populations consistent with the data was more useful.

I began with simulated annealing over synthetic populations. The loss matched the sorted round-robin score distribution on the leaderboard and the scores of known probe allocations, under several allocation priors. Different good fits could still imply different best replies.[4]

The marginal relaxation

The first inverse LP used $5 \times 101 = 505$ marginal probability masses. Each castle's masses sum to one; all are nonnegative; the expected soldier total is 100. Each probe score adds a linear band around its observed value. This is only a relaxation: marginal means summing to 100 do not guarantee a joint distribution supported on legal allocations.

The joint inverse problem

The stronger model uses mixture weights w_x on full legal allocations $x \in \mathcal{A}$. Define the probe-versus-column payoff

$$A_{jx} = \sum_{c=1}^{5} c \left[\mathbf{1}\{a_c^{(j)} > x_c\} + \frac{1}{2}\mathbf{1}\{a_c^{(j)} = x_c\} \right]. \tag{3}$$

For observed scores r_j, rounding allowances ϵ_j, and nonnegative residuals s_j, I solve

$$\begin{aligned} \min_{w,s} \quad & \sum_j s_j \\ \text{subject to} \quad & \left| \sum_{x \in \mathcal{A}} A_{jx} w_x - r_j \right| \le \epsilon_j + s_j, \\ & w_x \ge 0, \quad s_j \ge 0, \quad \sum_{x \in \mathcal{A}} w_x = 1. \end{aligned} \tag{4}$$

Later observations commonly used $\epsilon_j = 0.0005$; older, coarser probes had wider allowances. I first minimized total residual slack. I then minimized and maximized a candidate's score while allowing total slack up to that minimum plus a small numerical allowance. This does not force rounded, drifting measurements into false exact consistency.[3,5]

Why column generation mattered

There are $\binom{104}{4} = 4{,}598{,}126$ legal splits. I did not enumerate all of them in the LP. I solved a restricted problem, read its dual prices, searched for a full allocation with negative reduced cost, added that column, and repeated. The pricing search is exact dynamic programming because the dual-weighted objective is additive across castles.

Bounds are conditional on the observations, component model, slack budget, and pricing convergence. They are not unconditional guarantees about the real field.

3

What I could infer about the field

The honest output was a range of possible CDFs—not a reconstructed list of everyone's bids.

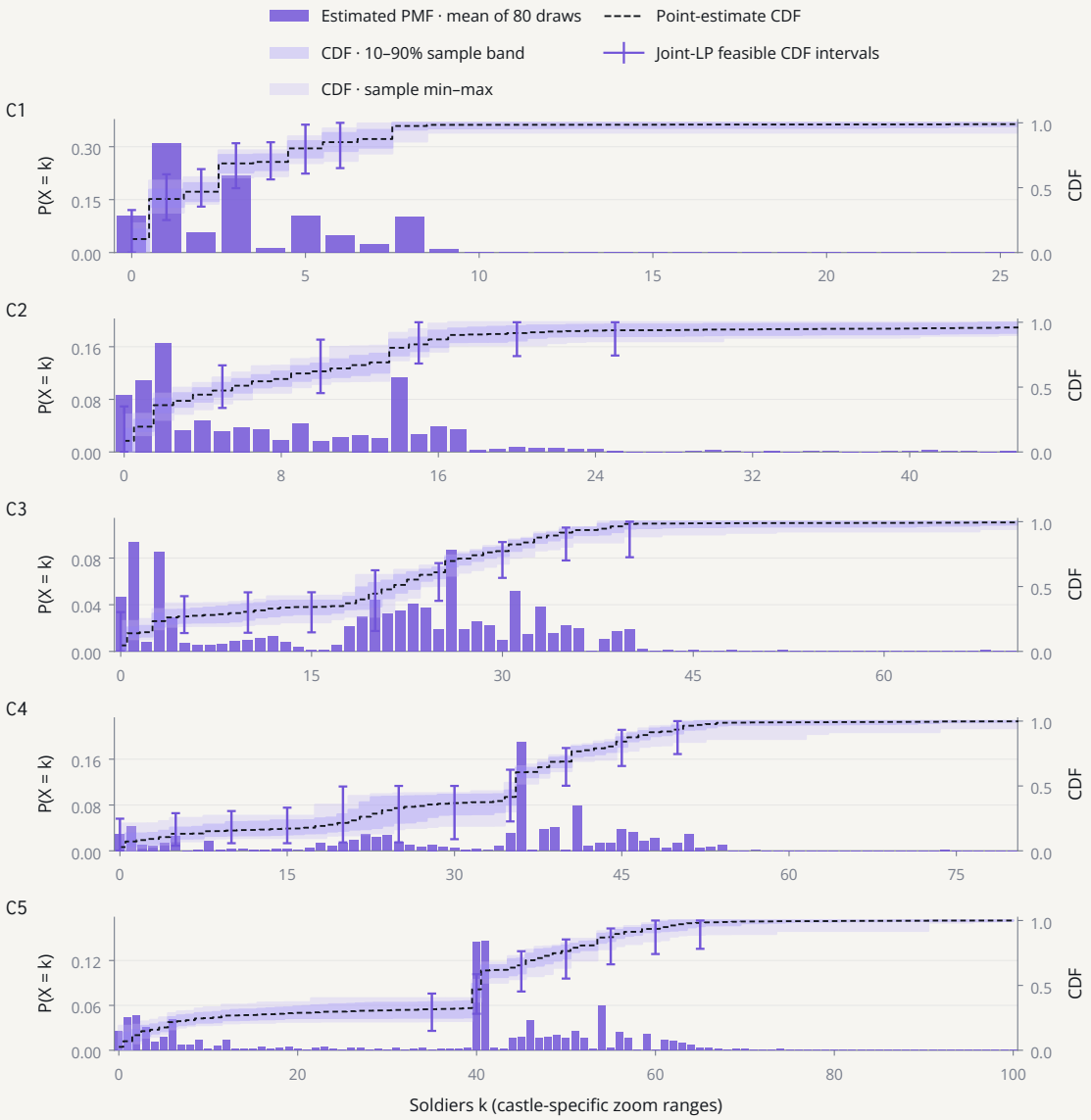


FIGURE 1a Final saved fit: r91, 114 score constraints through #113; refit recorded Nov 2, 2026, approximately 1:52 PM PST. Bars are the mean marginal PMF of 80 feasible-field draws; the dashed line and shaded bands use the CDF axis. Vertical whiskers are joint-LP feasible CDF intervals. Castle-specific zooms expose the mass. Source: r91/mix.pkl.

The end-state model is the latest completed fit with the most observations, not an intermediate reconstruction. Its 114 constraints include early probes and passive re-scores; this is not 114 independent experiments, nor the label of the excluded anomalous submission #114. The model describes the 4,238-entry field at that fit date, not the later final board.[5]

How uncertainty changed across trial stages

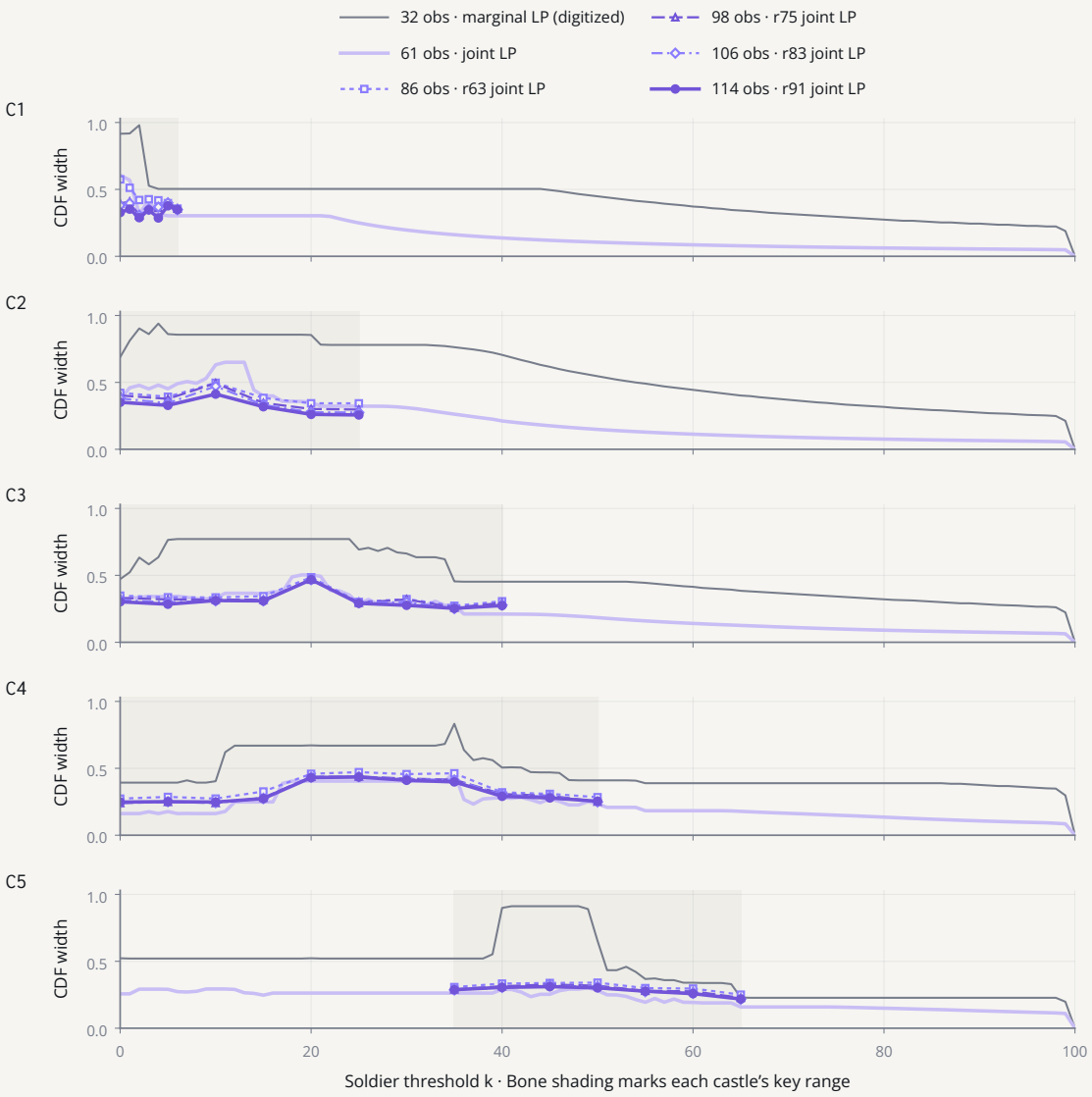


FIGURE 1b CDF interval width (upper minus lower feasible bound), all five castles: 32, 61, 86, 98, 106, and 114 score constraints. The 32-observation marginal-LP baseline is digitized from the historical chart; the 61-observation joint-LP curve uses saved full-grid bounds. Later stages use saved six-component joint-LP bounds only at marked thresholds; connecting segments are guides, not additional solved extrema. Bone shading marks the key ranges.

The stages use different field mixtures and model structures, so this is not a nested stationary-population contraction experiment. More observations need not tighten every threshold: new entrants, changed mixture weights, and slack can widen the feasible set.

The sample bands in Figure 1a summarize a heuristic ensemble, not Bayesian credible intervals or LP certificates. The LP whiskers optimize each CDF threshold separately; their endpoints may require different fields. For example, the final stored castle-1 bound is $F_1(3) \in [0.496, 0.843]$ after rounding. Enough information to choose an allocation can coexist with substantial uncertainty about the full field.

Sources: historical charts/w32_digitized.npy; charts/jband_tight.npy; and saved r63, r75, r83, r91 CDF extrema. Bounds are conditional on each model; no fresh public-board data are used.

4

One dynamic program, two jobs

The allocation space was large. Its budget structure made the optimization small.

For a fitted field, define the value of assigning k soldiers to castle c by equation (2):

$$v_c(k) = c\left[F_c(k-1) + \tfrac{1}{2}P_c(k)\right]. \tag{5}$$

The exact best reply follows a budget recurrence:

$$D_c(b) = \max_{0 \le k \le b} \{D_{c-1}(b-k) + v_c(k)\}. \tag{6}$$

Initialize $D_0(0) = 0$ and other initial budgets as infeasible; backtrack from $D_5(100)$. The running time is $O(5 \times 100^2)$. For LP pricing, replace the value table by the dual-derived reduced-cost table and minimize instead. The same routine optimizes my split and searches the huge space of opponent columns.[3]

Learning which field to trust

I optimized random linear objectives over the near-minimum-slack feasible set, producing extreme feasible fields. Later decision scripts mixed three such fields using Dirichlet weights, eventually with concentration 0.1. Convex mixtures preserve the linear constraints.

These were called “posterior samples” in the experiment, but they were a heuristic feasible-field ensemble, not an exact Bayesian posterior or a uniform polytope sample. I computed each sampled field’s best reply and added nearby legal allocations to form an approximate possibly-optimal candidate set.

$$R(a,p) = \max_{b \in \mathcal{A}} S_p(b) - S_p(a), \qquad \mathrm{MMR}(\mathcal{P}) = \min_{a} \max_{p \in \mathcal{P}} R(a,p). \tag{7}$$

For each potential probe q, I split sampled fields into three bins by their predicted score on q. I recomputed minimax regret inside each bin and weighted by the bin’s sample frequency. The probe with the largest expected regret reduction was most useful; near ties favored higher expected score.[5]

Two different kinds of uncertainty. LP score extrema explored the feasible set. Regret calculations used sampled fields and a finite candidate set. The latter were decision heuristics, not certified worst-case regret over every feasible field.

5

Experiments, not random hill climbing

Some of the most useful submissions made my interim rank worse. That was part of the design.

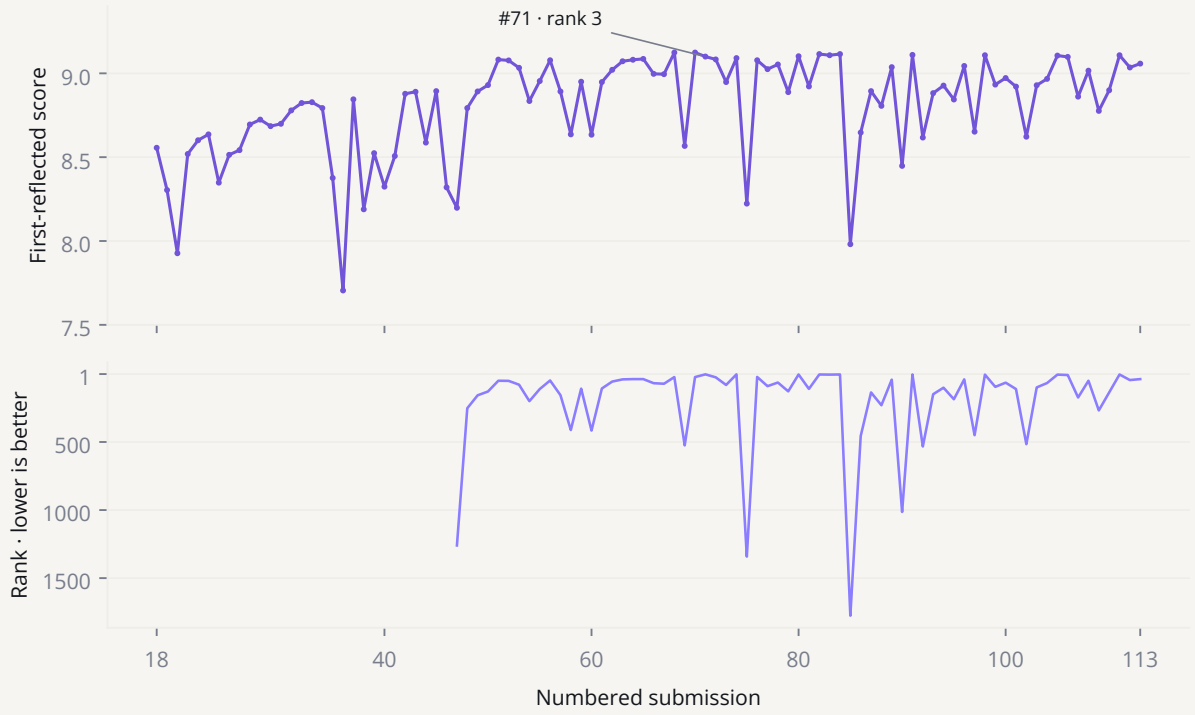


FIGURE 2 First-reflected scores for #18--#113; ranks begin at #47. The horizontal axis is numbered submission, not elapsed time. Scores span changing fields. Suspect #114 and unreflected submissions are excluded. Source: reconciled submission log and observations.

The 96 score observations are deliberately non-monotonic. Low-scoring experiments can eliminate plausible fields that disagree about the best final decision. The rank view makes the short-term cost visible; it should not be mistaken for a monotonic optimization trace.

A probe that also had competitive upside

Submission #71, $(4, 6, 35, 48, 7)$, offered an estimated regret reduction of 0.052 versus the top probe's 0.054. It was also the ensemble's best lower-decile choice. I chose it as a useful experiment with a strong expected downside, not merely because its mean looked attractive.

It returned 9.100 and rank 3. A large entrant wave arrived in that same refresh, so comparing its outcome directly with the pre-wave prediction would not be a clean calibration test. Subsequent re-tests helped establish that the allocation remained competitive.[1,5]

6

Calibration was part of the algorithm

A narrow ensemble is not useful if reality repeatedly lands outside it.

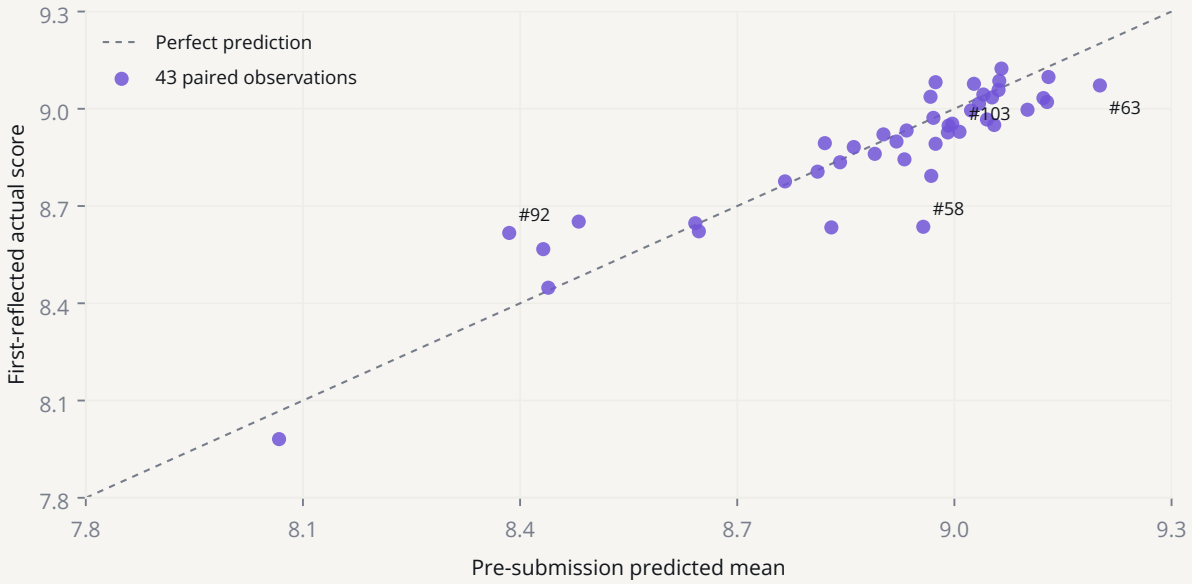


FIGURE 3 43 explicit pre-submission predicted means paired with first-reflected scores. Rows come from #48--#70 and #85--#113, excluding drift re-tests and rows without an explicit mean in the purpose column. The dashed line is equality. Source: submission log.

I tracked both prediction error and the percentile of the actual score within the ensemble. Submission #62 landed near the first percentile; #63 fell below its predicted range. That exposed an optimistic sampler. I reduced weight on the optimistic central fit and pushed mixtures toward extreme feasible fields instead of treating the mean as ground truth.

Later, #103 and #104 each scored 0.078 below their predicted means; #106 landed at the sample minimum. Several ordinary mid-range outcomes followed in #107–#113. The response was a loop: adjust the sampler, re-test a known split, and distinguish model bias from genuine field drift.[1,5]

A diagnostic, not a benchmark. These are adaptive experiments, not held-out observations. I omit the intervening field-break period because a prediction for one population is not a clean test on another. The plotted subset is defined by the availability of explicit logged means, not a claim of complete coverage.

7

The population was moving

An unchanged allocation can earn a different score when its opponents change.

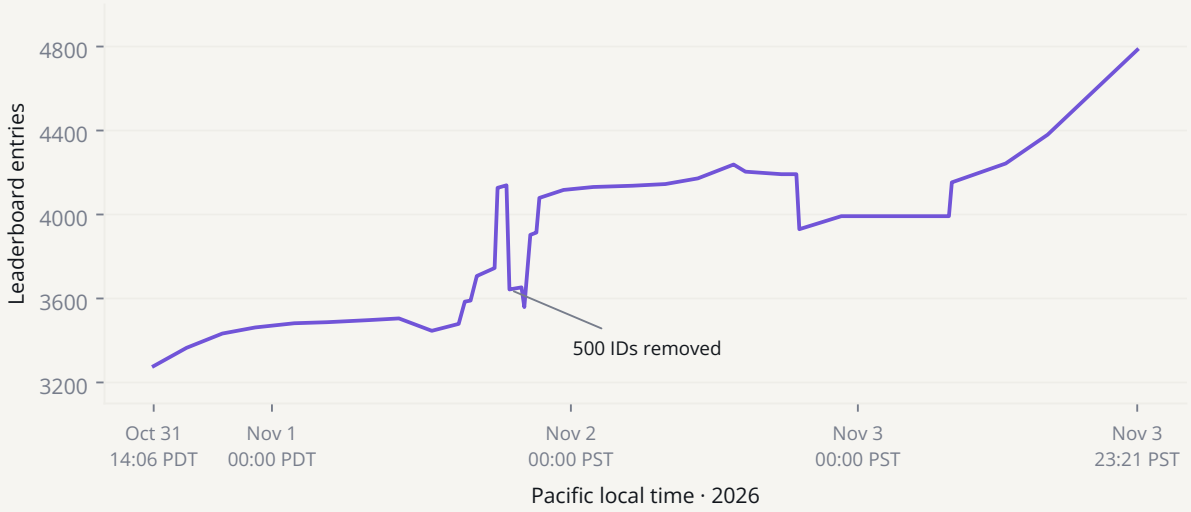


FIGURE 4 38 representative board readings, preserving large jumps, selected from 175 unique saved refresh timestamps. Lines join observations; they do not imply continuous ingestion. Source: saved leaderboard JSON snapshots.

On Nov 1, the field went from 3,479 to 3,585 entries in one observed refresh. Later it went from 3,745 to 4,127, then from 4,139 to 3,643 as 500 IDs disappeared and four arrived. Treating every score as one stationary population would manufacture contradictions.[1,2]

I modeled entry waves as separate unknown allocation distributions:

$$p_t = \sum_h \alpha_{t,h} p_h, \qquad \alpha_{t,h} = \frac{N_{t,h}}{N_t}. \tag{8}$$

Each measurement constrained the mixture present at its own timestamp. Joins and removals changed weights; a removed cohort could still explain historical scores without dominating the current field. The final pre-freeze model had six components.[5]

Repeated splits served as anchors. The allocation $(6, 6, 35, 46, 7)$ scored 9.124 before the first large wave and 9.083 afterward. A re-test measures drift at a known point; it should not automatically be counted as another independent constraint. Component membership and composition remained approximations, so I retained slack and monitored residuals.

Time labels are Pacific local time. Daylight saving ended on Nov 1, 2026; the figure labels distinguish PDT and PST.

8

What I would keep—and change

The reusable result is the measurement-to-decision loop, not a magic allocation.

1. **Keep the loop explicit.** Submit a deliberate probe; wait for a genuinely ingesting refresh; reconcile identity and score; update the feasible model; calibrate; choose the next experiment.
2. **Treat the pipeline as part of the system.** An advancing board timestamp did not guarantee ingestion. On Nov 3, other entries joined while my existing entries were not replaced. That is an observed symptom, not a proven server-side diagnosis.
3. **Separate receipt, reflection, and prediction.** A thank-you screen established receipt only. Freeze-time submissions remained unreflected and were operationally treated as dropped. Final leaderboard replacement was not verified.
4. **Validate inputs as strictly as the model.** I checked castle order and the 100-soldier total, captured confirmations, and kept a backed-up append-only log. I excluded anomalous #114 from the LP because a possible form-entry error was suspected; that diagnosis was never proven.
5. **Prefer validated downside to attractive extrapolation.** Near the deadline I restored the best-tested split. Next time I would reserve more time for ingestion checks and keep feasible bounds visibly separate from ensemble percentiles.

Evidence and limitations

The final log label is #126, not a count of independent measurements. Some early rows were reconstructed after a truncation, and #35/#36 allocations were inferred. The machine-readable observations include repeated labels, pending submissions, and passive re-scores; I did not treat every row as a new trial. Historical marginal pictures were estimates from other cohorts, not raw bid ground truth.

This retrospective uses saved boards through Nov 3, 11:21:52 PM PST. It makes no claim about eventual official standings. The CDF intervals, sampled regret, and fitted component model each carry different assumptions.

Sources and reproducibility

[1] `submissions.md`; `observations.json`. Numbered experiments, first-reflected scores, predictions, re-tests, anomalies, and receipt records.

[2] Saved `lb_*.json` snapshots. Field counts, identity continuity, score drift, and last observed ranking. Public endpoint: pokerbot.sequoiacap.com/castle_challenge/data/leaderboard.json.

[3] `lp_inverse/cg.py`. Payoff kernel, minimum-slack LP, column pricing, and exact best reply.

[4] `fit.py`; `analyze.py`; `lpbounds.py`. Synthetic populations, marginal relaxation, and optimization.

[5] `r91/mix6.py`; `r91/dec.py`; `r91/mix.pkl`. Six-component model, CDF extrema, Dirichlet mixtures, and sampled regret selection.

Figures 1a--b are reproduced by build_game1_detail_charts.py from saved fits; Figures 2--4 by build_game1_charts.py from chart_data_game1.json, with vector PDF/SVG and PNG outputs. Font substitutes: Gelasio (editorial), Open Sans (body), Inconsolata (labels); mathematical notation uses Computer Modern.